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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Comparing each element with the others
- c. Combining sorted sublists
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Quick Sort
- b. Merge Sort
- c. Insertion Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Sorting the entire list in one pass
b. Dividing the list into halves
c. Bubbling up the largest element to its correct position
d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **4**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It sorts data out of the box efficiently
b. It only works with integer arrays
c. It requires external libraries
d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search
b. Bubble Sort
c. Merge Sort
d. Binary Search

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It divides the list into sublists
- c. It recursively sorts subproblems
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **7**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Selection
- b. Extraction
- c. Distribution
- d. Insertion

The correct answer is:
Insertion

Question **8**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Searching
- b. Merging
- c. Complexity
- d. Sorting

The correct answer is:
Complexity

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Quick Sort
c. Bubble Sort
d. Heap Sort

The correct answer is: Bubble Sort

Question **10**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Insertion sort
b. Quick sort
c. Heap sort
d. Bubble sort

The correct answer is:
Insertion sort

Question **11**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By comparing elements sequentially
c. By sorting data in a single pass
d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **12**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed
b. It makes the search operation slower
c. It allows for faster searching
d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **13**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. order()
b. sorted()
c. arrange()
d. sort()

The correct answer is: sorted()

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Bubble Sort
b. Merge Sort
c. Binary Search
d. Linear Search

The correct answer is: Bubble Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort is difficult to implement
c. Bubble Sort is less efficient for large lists
d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists